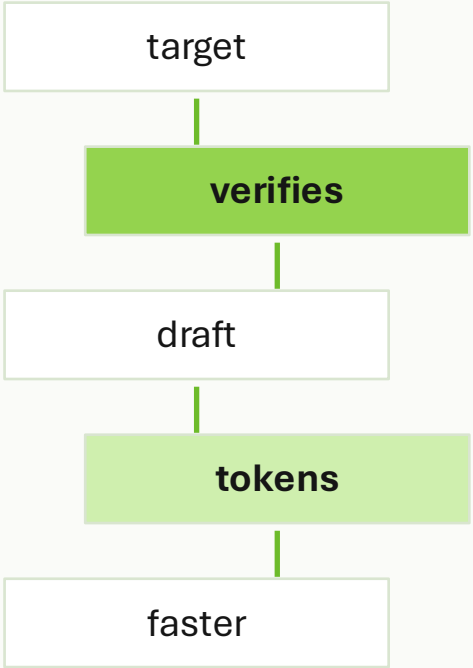
Speculative Decoding Acceleration for LLMs

Benchmarking Gemma Speculative Decoding and EAGLE-3 on Mid-Tier GPU Hardware



Core idea

Use a draft model or draft head to propose candidate tokens, then verify them with the target model.

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Problem Selection

Why speculative decoding is a meaningful NLP systems problem

LLMs generate text autoregressively: one token at a time.

Each new token requires another decoding step, so inference can become slow and memory-bandwidth bound.

Before		After
one token / pass	→	several tokens / step

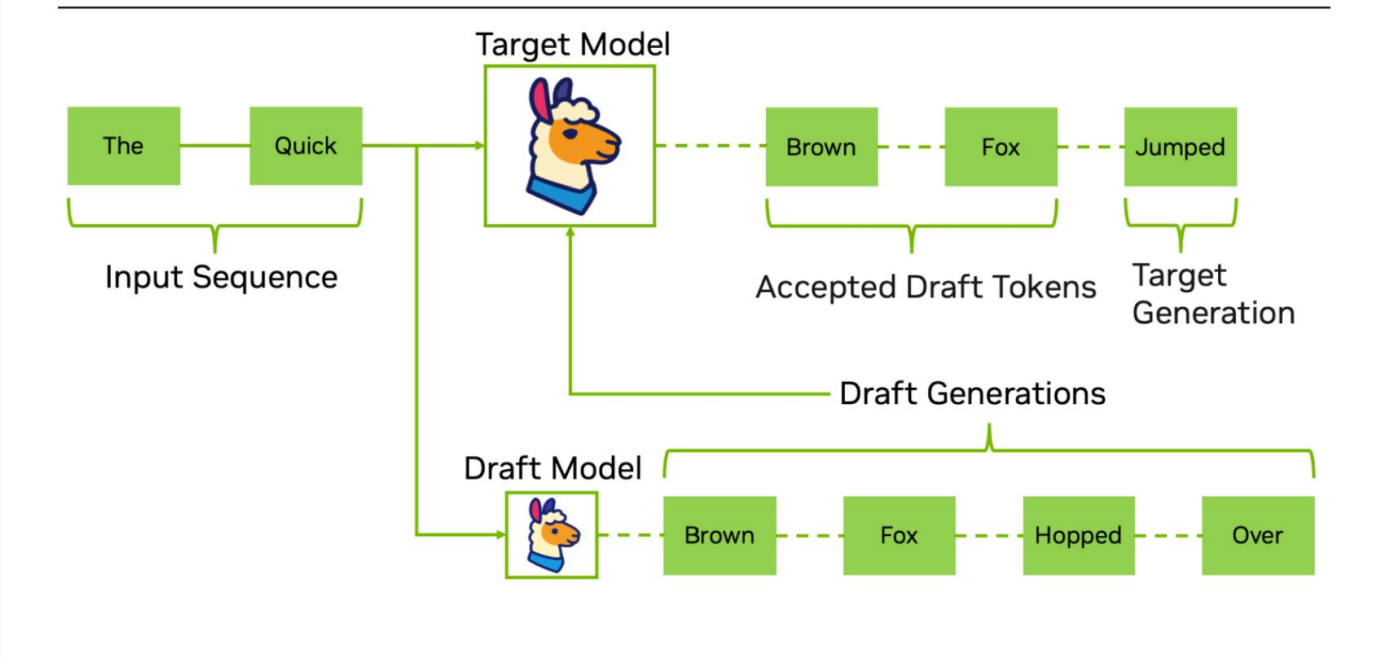
We selected this because:

- It addresses a real inference bottleneck in autoregressive LLM generation.
- Most benchmarks use flagship GPUs, leaving less guidance for A10G-style hardware.
- It connects core NLP concepts: autoregressive decoding, probability distributions, and sampling.

Speculative decoding uses a draft component to propose candidate tokens, and the target model verifies them. If the candidates are accepted, generation can advance by multiple tokens in one step.

How Standard Speculative Decoding Works

Draft first, verify once, then continue from the accepted prefix



Step-by-step logic

1 Input context

Both models start from the current prefix.

2 Draft proposal

The draft model proposes γ candidate tokens.

3 Parallel verification

The target evaluates the proposal in one batched step.

4 Accept or resample

Accepted tokens are appended; the process repeats.

Key variable: γ

γ controls how many draft tokens are proposed before verification.

Final Gemma Experimental Scope

Final report focuses on two completed acceleration modules for Gemma-3-12B

Final configurations

Pair	Method	Target	Draft / Head	VRAM
F	Standard	gemma-3-12B	gemma-3-1B	< 10GB
I	EAGLE-3	gemma-3-12B	trained draft head	< 10GB

Hardware target **AWS g5.2xlarge · NVIDIA A10G · 24 GB VRAM**

Why the shift matters

- Early model-selection trials narrowed the scope. The final report focuses on completed Gemma-3-12B results: Pair F standard speculative decoding and EAGLE-3 draft-head acceleration.
This creates a clear comparison between a separate draft model and a learned draft head.

Module 1: Standard speculative decoding

Pair F uses Gemma-3-12B-it as the target and Gemma-3-1B-it as the separate draft model. The module tests whether a smaller Gemma draft can improve throughput for the larger target.

Module 2: EAGLE-3 draft head

EAGLE-3 replaces the full separate draft model with a learned draft head operating on the target model's hidden states. This tests a more compact acceleration path.

Datasets

Four task domains test when speculative decoding helps or hurts

We evaluate acceptance behavior across prompt sets with different output predictability.

HumanEval

Code completion

50 prompts

highly structured

Trivia QA

Question answering

50 prompts

knowledge-focused

CNN/DailyMail

Summarization

50 prompts

moderately constrained

WritingPrompts

Creative writing

50 prompts

high entropy

Structured output → **High-entropy output**

All prompt sets use 50 prompts sampled with seed 42 from Hugging Face Datasets.

Module 1: Standard Speculative Decoding

We vary speculation length and randomness across all tasks

Control variables

γ values	1	3	5	7	10
temperature	0.0	0.6	1.0		
tasks	HumanEval	TriviaQA	CNN/DM	Writing	

Active configuration

Pair	Target	Draft	Quantization
F	Gemma-3-12B-it	Gemma-3-1B-it	4-bit target; BF16 draft

Run flow

load prompts
4 task x 50 prompts

↓

run baseline
target only; no draft tokens

↓

run speculative
γ × temperature × task

↓

summarize CSV

↓

visualize comparison

Metrics captured

- acceptance rate
- accepted length
- speedup
- TTFT
- peak VRAM
- draft overhead
- TPS / throughput

Goal

Find when the speed gained from accepted draft tokens outweighs the extra draft-model cost.

Model Setup

We vary speculation length and randomness across all tasks

Setup overview

- Benchmark runtime: around 10 hours
- Pair F: target = Gemma-3-12B-it, draft = Gemma-3-1Bit
- Bothe models come from the same Gemma instruction – tuned family
- Similar model family helps align token distributions between draft and target

Model Comparison: Pair F

Target Model

Gemma-3-12B-it

Parameters:	~12B parameters
Precision:	4-bit quantized
Note:	Lower GPU memory usage

Draft Model

Gemma-3-1B-it

Parameters:	~1B parameters
Precision:	BF16
Note:	stable and efficient generation

4-bit quantization stores each weight with fewer bits, which reduces GPU memory usage while keeping the same model size in parameters.

GOAL: Keep the target model strong, fit the setup on available GPU memory, and use a smaller draft model for efficient candidate generation

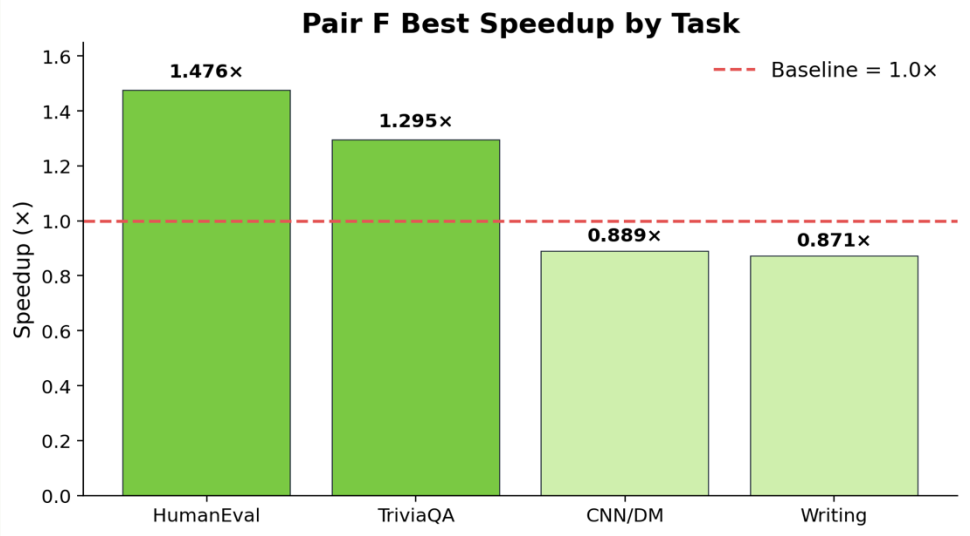
Pair F Results: Speedup Is Task-Dependent

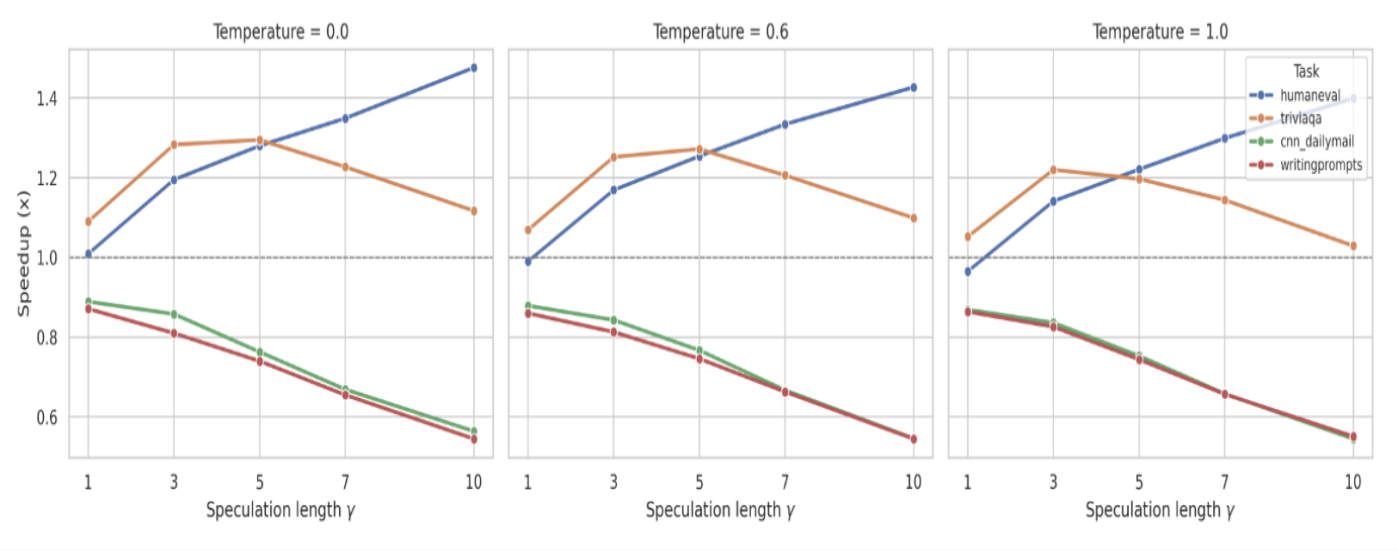
Structured tasks exceed baseline, while open-ended tasks do not

Task	Best gamma	Best temperature	Mean TPS	Speedup	Mean Acceptance Rate	Mean Acceptance Length	Mean TTFT (ms)	Mean Peak VRAM (GB)
humaneval	10	0.0	14.22	1.476	0.9279	6.03	555.24	9.40
triviaqa	5	0.0	12.47	1.295	0.8751	2.96	320.48	9.31
cnn_dailymail	1	0.0	8.30	0.889	0.5342	0.53	525.01	9.77
writingprompts	1	0.0	8.43	0.871	0.4525	0.45	180.03	9.32

Takeaway

- HumanEval and TriviaQA exceed baseline, while CNN/DailyMail and WritingPrompts fall below baseline.
- Increasing γ does not always improve speedup; it helps only when the extra drafted tokens are actually accepted.





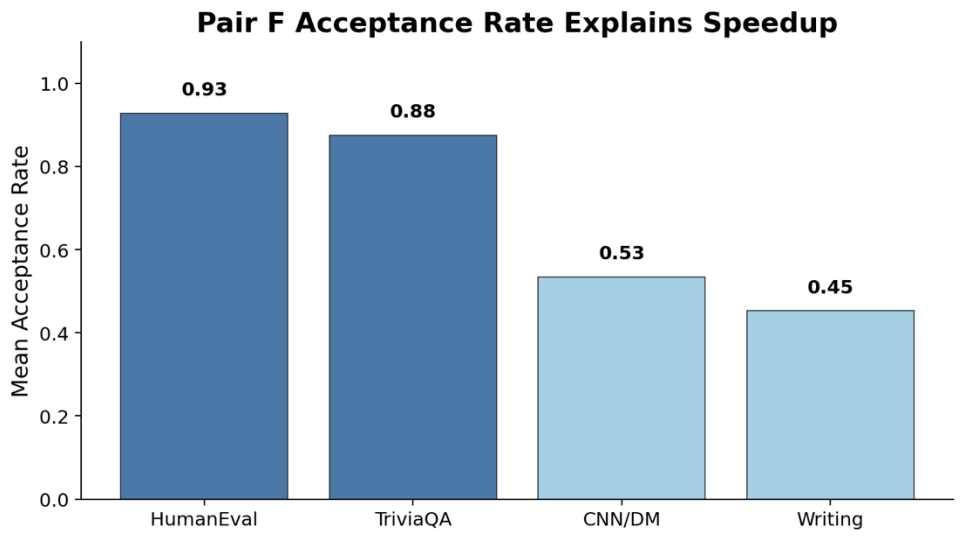
Pair F Observation: Acceptance Explains Speedup

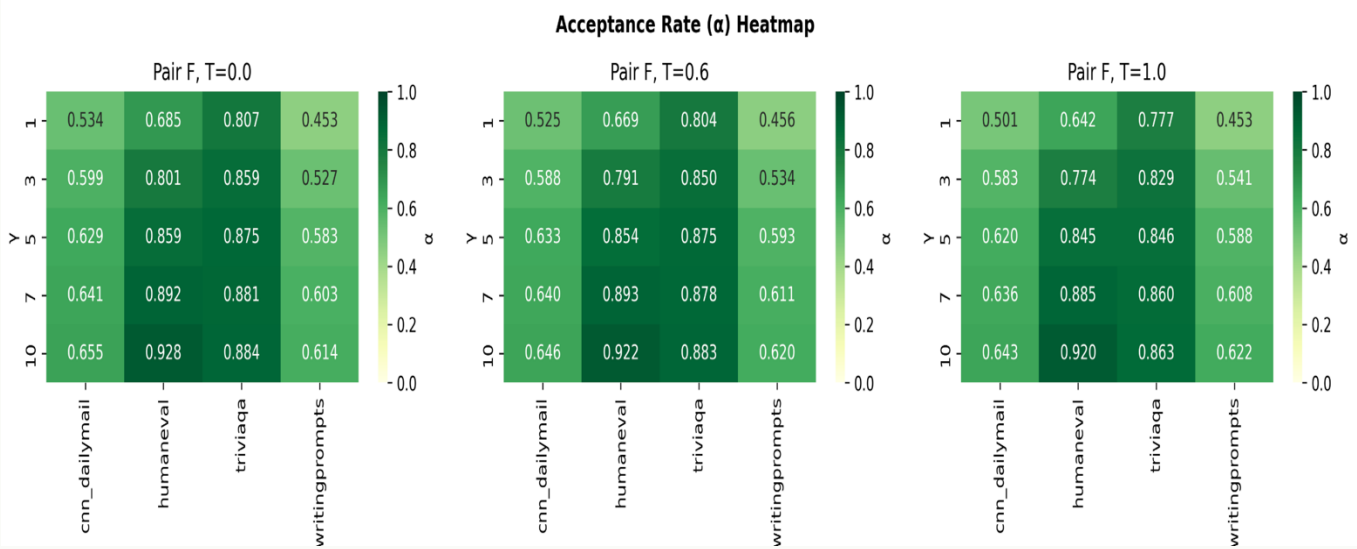
Higher draft-target agreement produces stronger throughput gains

Task	Best `gamma`	Best `temperature`	Mean TPS	Speedup	Mean Acceptance Rate	Mean Acceptance Length	Mean TTFT (ms)	Mean Peak VRAM (GB)
`humaneval`	10	0.0	14.22	1.476	0.9279	6.03	555.24	9.40
`triviaqa`	5	0.0	12.47	1.295	0.8751	2.96	320.48	9.31
`cnn_dailymail`	1	0.0	8.30	0.889	0.5342	0.53	525.01	9.77
`writingprompts`	1	0.0	8.43	0.871	0.4525	0.45	180.03	9.32

Takeaway

- HumanEval and TriviaQA maintain higher acceptance rates, while CNN/DailyMail and WritingPrompts remain lower.
- When acceptance is low, many draft tokens are rejected, so extra draft-model work can make decoding slower instead of faster.





Pair F Observation: Throughput–Latency Trade-off

Best speculative settings can improve throughput but increase TTFT

Baseline

Task	Baseline TPS	Baseline TTFT (ms)	Baseline Peak VRAM
humaneval	9.63	179.12	9.32
triviaqa	9.63	134.99	9.26
cnn_dailymail	9.34	462.10	9.75
writingprompts	9.68	142.32	9.27

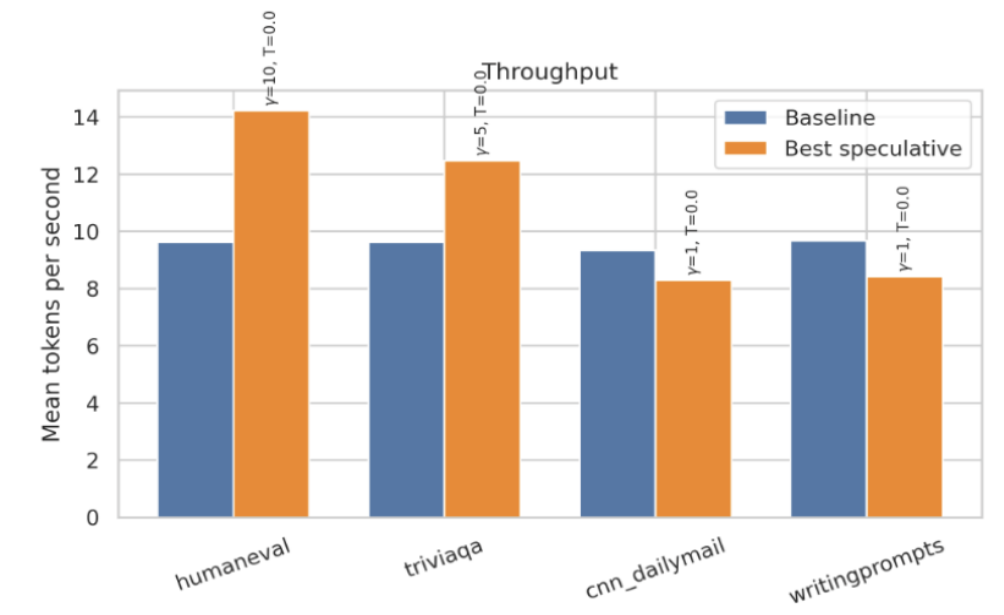
Speculative

Task	Best gamma	Best temperature	Mean TTFT (ms)	Mean Peak VRAM (GB)
humaneval	10	0.0	555.24	9.40
triviaqa	5	0.0	320.48	9.31
cnn_dailymail	1	0.0	525.01	9.77
writingprompts	1	0.0	180.03	9.32

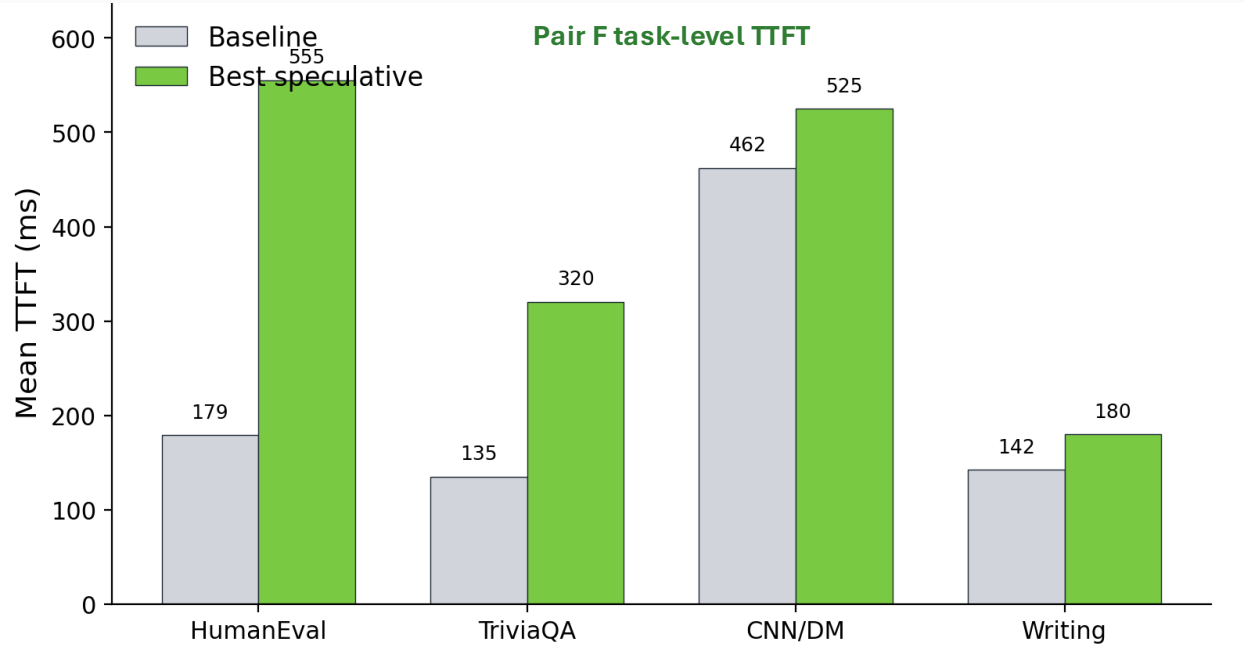
Takeaway

- Pair F can improve sustained throughput on structured tasks, but the best speedup settings can substantially increase TTFT.

Pair F task-level throughput



Pair F task-level TTFT

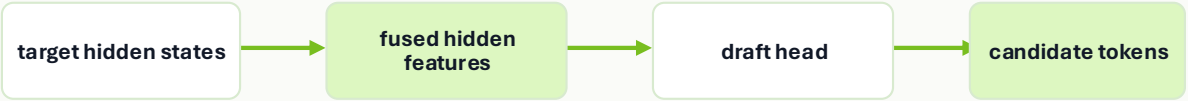


Module 2: EAGLE-3 Draft Head

A learned draft head built on Gemma target hidden states

Active configuration

Pair	Target	Draft Component	Quantization
I	Gemma-3-12B-it	trained EAGLE-3 draft head	4-bit frozen target



Objective

- Replace the separate full draft model with a learned draft head.
- Train a draft head that predicts future tokens from target hidden states.
- Use a 4-bit frozen Gemma-3-12B target during training and inference.

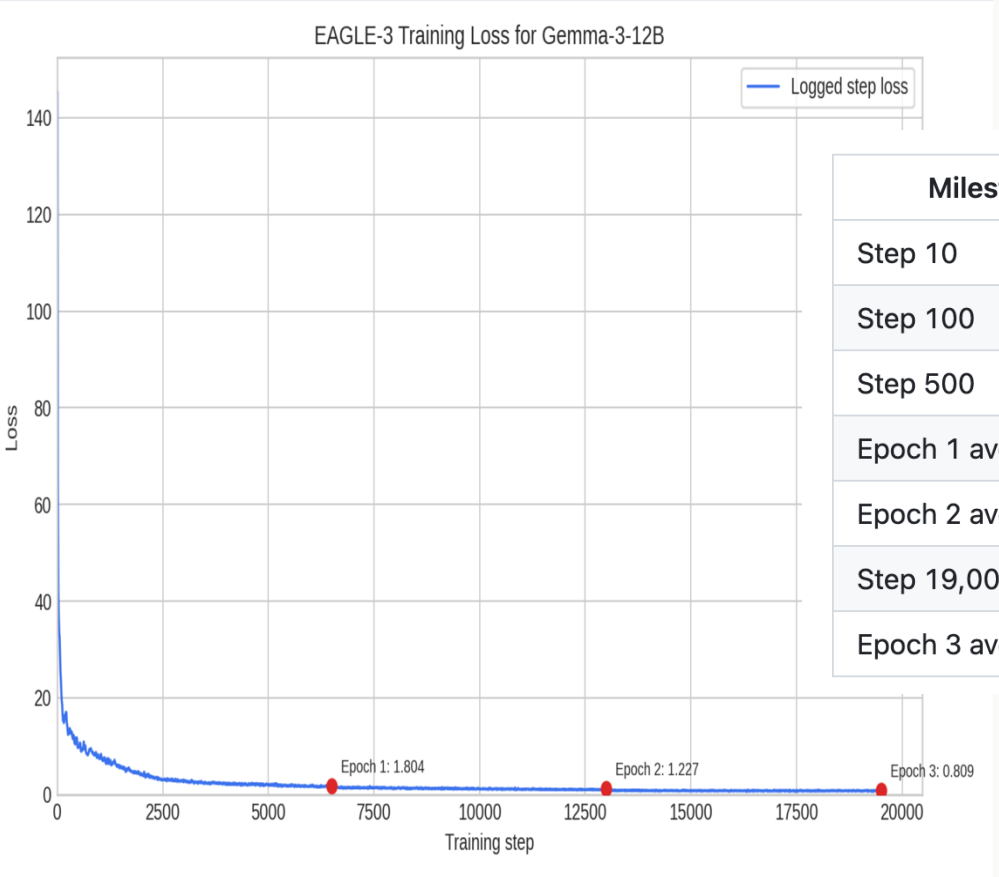
Model ingredients

- three target hidden layers
- fusion + input projections
- target embedding layer
- one copied decoder layer
- frozen norm + LM head

Pair I : EAGLE-3 Training Results

Module 2: EAGLE-3 Draft Head Training Draft Head

Item	Observed Value
Logged target model	`google/gemma-3-12b-it`
Logged hardware	`NVIDIA A10G`
Prepared examples	`52,002`
Trainable parameters	`297,876,992`
Optimizer	8-bit AdamW
Final checkpoint	`eagle3_gemma3_12b_final.pt`
Training completion time	`2026-04-27 18:13:37`



Milestone	Loss
Step 10	145.2908
Step 100	21.2564
Step 500	9.7899
Epoch 1 average loss	1.8044
Epoch 2 average loss	1.2272
Step 19,000	0.7941
Epoch 3 average loss	0.8095

Takeaway

- Training completed successfully: loss dropped from 145.3 at step 10 to below 1.0 by the final epoch, producing a usable EAGLE-3 checkpoint.

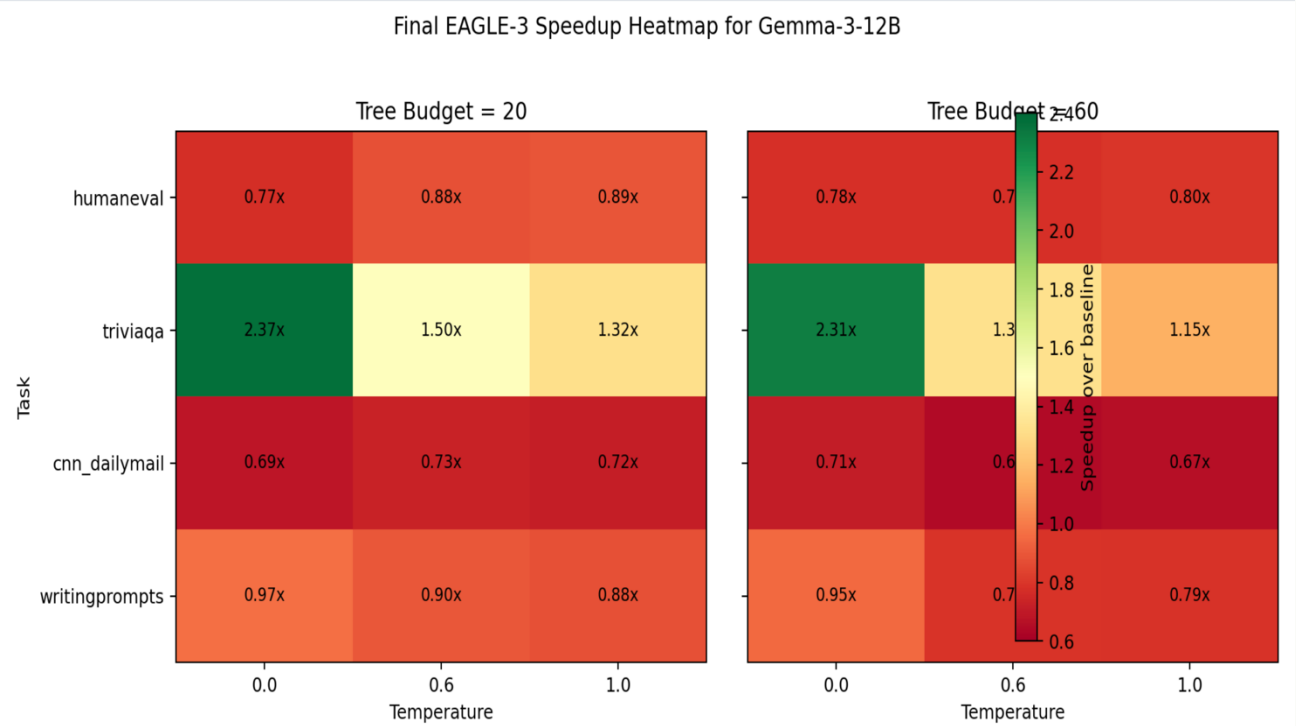
Pair I Results: EAGLE-3 Helps Selectively

Module 2: EAGLE-3 Draft Head

Task	Best Tree Budget	Best Temperature	Mean TPS	Speedup	Mean Acceptance Rate	Mean TTFT (ms)	Mean Peak VRAM (GB)
humaneval	20	1.0	7.61	0.893	0.0974	202.53	8.29
triviaqa	20	0.0	20.87	2.368	0.8977	162.60	8.11
cnn_dailymail	20	0.6	6.03	0.727	0.0803	476.55	9.25
writingprompts	20	0.0	8.01	0.969	0.2206	169.72	8.39

Takeaway

- TriviaQA is the strongest EAGLE task by far, peaking at 2.368x speedup for tree budget = 20 and temperature = 0.0
- WritingPrompts gets close to baseline, but HumanEval and CNN/DailyMail remain below baseline.
- CNN/DailyMail remains the hardest case, suggesting long-context verification is still a limitation.



Conclusion

The final results show when Gemma inference acceleration helps in practice

Takeaways

1. Pair F and EAGLE-3 provide two completed paths for Gemma inference acceleration.
2. Speedup depends on draft-target agreement; low acceptance can eliminate the benefit.
3. Practical deployment requires balancing throughput, TTFT, VRAM, and engineering constraints.

Next step Improve long-context EAGLE verification for CNN/DailyMail-style tasks.

Thank you